

The Volatility of Inflation Expectations and Interest Rates

by **Bonaglia, d'Arienzo, Fallico, Gennaioli, and Iovino**

Discussion by Oliver Pfäuti (UT Austin)

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Summary

- ▶ Linking inflation expectations of banks in 18 advanced economies (1989–2022) to interest rates, paper documents **overreaction at longer horizons:**
 - ▶ high current inflation $\Leftrightarrow$ negative forecast errors and high real returns
- ▶ Develop and estimate **mnemonic learning** model **replicating empirical findings:**
 - $\Rightarrow$ **state- and history-dependent expectations** that vary with countries' inflation histories and characteristics (volatility & persistence)
- ▶ **Macroeconomic implications:** high expected inflation today predicts future transfers from borrowers to lenders, potentially amplifying booms & busts

General assessment

Different (differing?) views & findings on (inflation) expectations over last decade:

- ▶ Underreaction of short-horizon forecasts (e.g., Coibion/Gorodnichenko)
- ▶ Horizon-increasing overreaction (e.g., Afrouzi et al., Halperin/Mazlish, D'Arienzo)
- ▶ Overreaction to news and past states (e.g., Bordalo et al.)
- ▶ Term-structure of expectations (e.g., Angeletos et al.)
- ▶ Long-lasting effects of experiences (e.g., Malmendier/Nagel)
- ▶ State-dependent inflation expectations (e.g., Pfäuti), linked to people's experiences (e.g., Gennaioli et al., or Link et al.)

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Discussion:

1. Are banks special?
2. Rule out other mechanisms
3. Potential macroeconomic (monetary policy) implications of mnemonic learning

Comment 1: Are banks special?

- ▶ Why are banks' belief biases not priced in?
 - ▶ Do consensus forecasts differ from marginal investor's beliefs? Do they not act on those forecasts?
 - ▶ Are strategic considerations in forecasts a potential issue and do forecasts not reflect their true beliefs (see, e.g., Baley/Turen (2025))?

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- ▶ Banks and professional forecasters are also likely special when it comes to the cue that triggers memory recall: current inflation π_t
 - ▶ Why? Because they are likely very attentive to inflation, whereas households and firms systematically differ in how attentive they are to inflation

Attention, experiences and inflation expectations

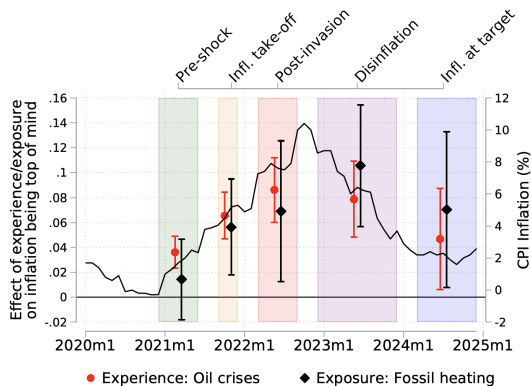
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	Households			
	(1)	(2)	(3)	(4)
	Expected inflation next 12 months	Abs. dev. from consensus forecast (Focus)	Abs. dev. from swap-based market expectations	Abs. dev. from AR(4) forecast
Experience: Oil crises				
× 1(Inflation take-off)	0.647*** (0.142)	0.627*** (0.140)	0.565*** (0.139)	0.482*** (0.137)
× 1(Post invasion)	1.050*** (0.174)	0.941*** (0.169)	0.655*** (0.162)	0.754*** (0.164)
× 1(Disinflation)	0.895*** (0.191)	0.795*** (0.186)	0.803*** (0.186)	0.809*** (0.185)
× 1(Inflation at target)	0.405** (0.200)	0.369* (0.196)	0.397** (0.195)	0.410** (0.194)

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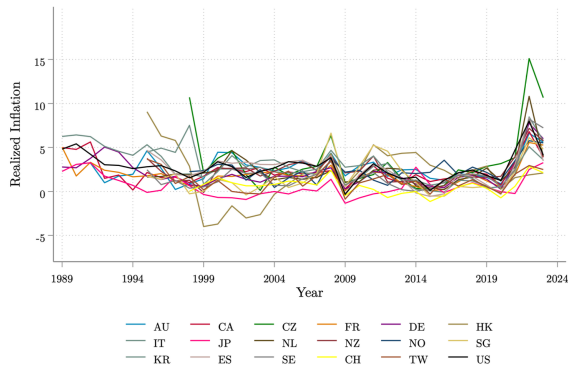
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- ⇒ Experience-based attention likely interacts with experience-based expectation formation

Comment 2: Global shocks and time-varying risk premia

Figure 6: Realized Inflation over the sample

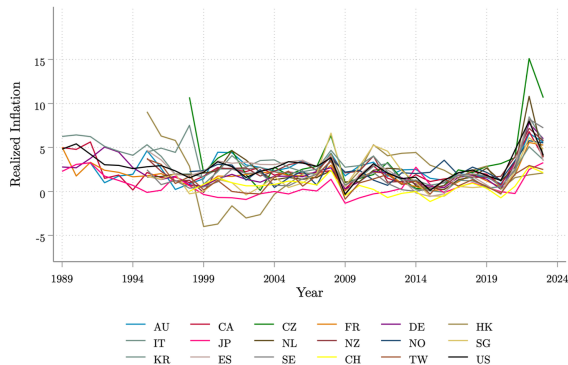


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Note. Annual realized inflation for the 18 countries in our sample, from 1989 to 2023. Sample varies according to Table 6. The identifier for each series corresponds to the ISO 3166-2 country abbreviation.

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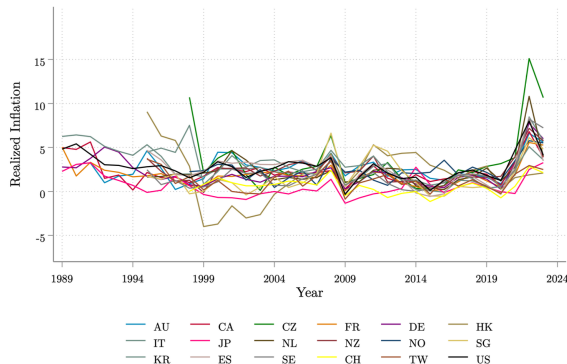


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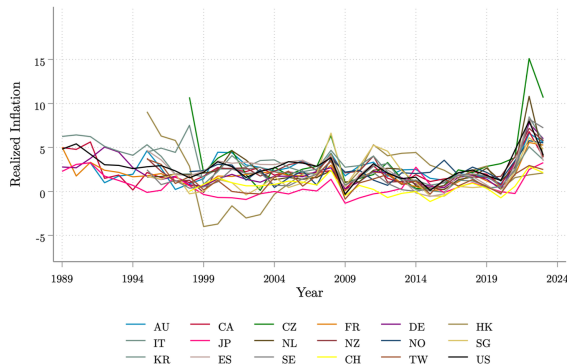


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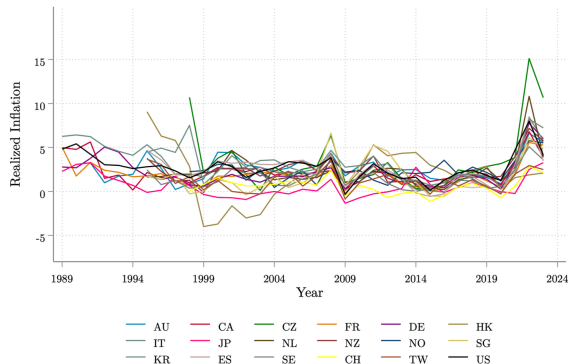


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 - ⇒ Since inflation-adjusted bonds are often not available, use proxies (inflation volatility, uncertainty, disagreement) + time FEs

Macro Implications

- ▶ How does mnemonic learning affect the conduct of monetary policy?
- ▶ Monetary policy leans against inflation: $i_t = \gamma_\pi \pi_t$.
- ▶ Static Phillips Curve: $\pi_t = \kappa x_t$.
- ▶ Demand shock ε_t^D with persistence $\rho \Rightarrow \mathbb{E}_t^{RE} \pi_{t+h} = \rho^h \pi_t$.
- ▶ Standard New Keynesian model but households form expectations as estimated in Bonaglia et al.:

$$\tilde{\mathbb{E}}_t \pi_{t+h} = \phi(\alpha, \beta, h) \pi_t$$

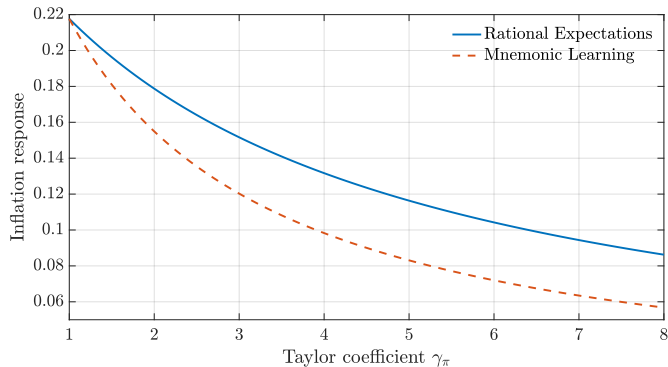
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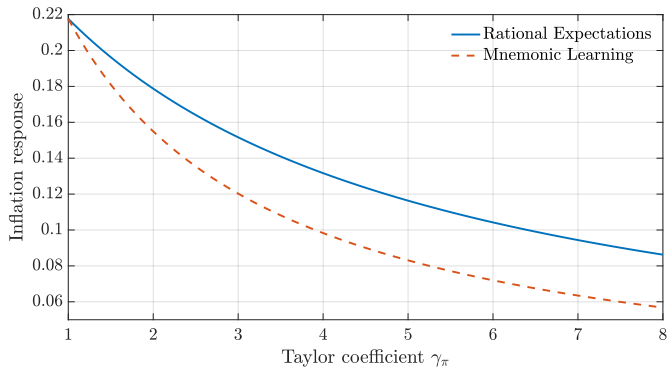
- ▶ How “effective” is monetary policy in reducing the inflationary pressures from demand shocks?
 - $\Rightarrow$ solve for **inflation response for different γ_π** under rational expectations, and under mnemonic learning

Monetary policy is more effective under mnemonic learning



- MP is more effective under mnemonic learning if $\gamma_\pi > 1$

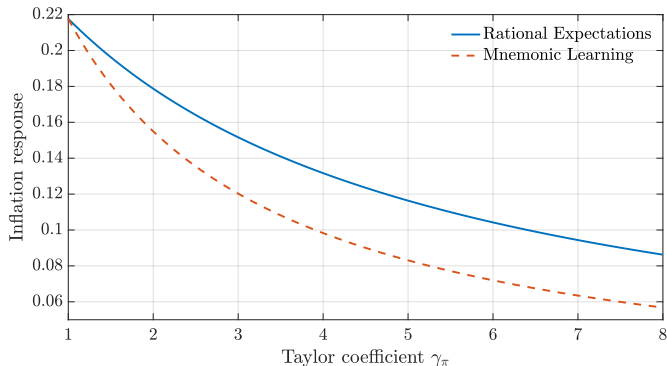
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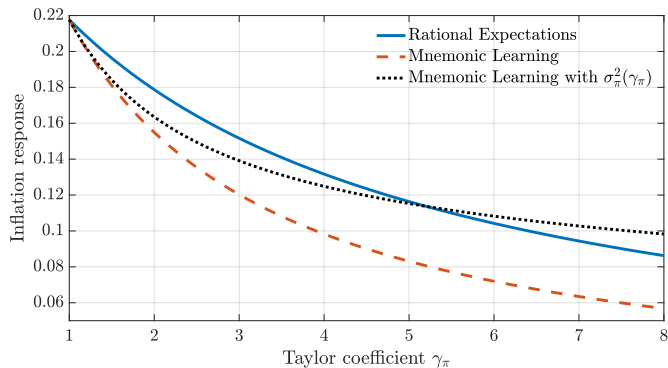


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⇒ what if more hawkish MP reduces σ_π^2 and through that affects $\phi(\alpha, \beta, h)$?

Monetary policy effectiveness when experiences are endogenous



⇒ monetary policy can become **less effective** when it “kills” people’s overreaction

Concluding remarks

- ▶ Beautiful paper whose implications open the door to many questions while providing an elegant framework to tackle these questions:
 - ▶ Macroeconomic implications potentially very rich and far from obvious
 - ▶ Other potentially far-reaching implications: asset pricing, household finance (borrowers vs saver, ...), macro-finance (default, financing structure, ...), etc.
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